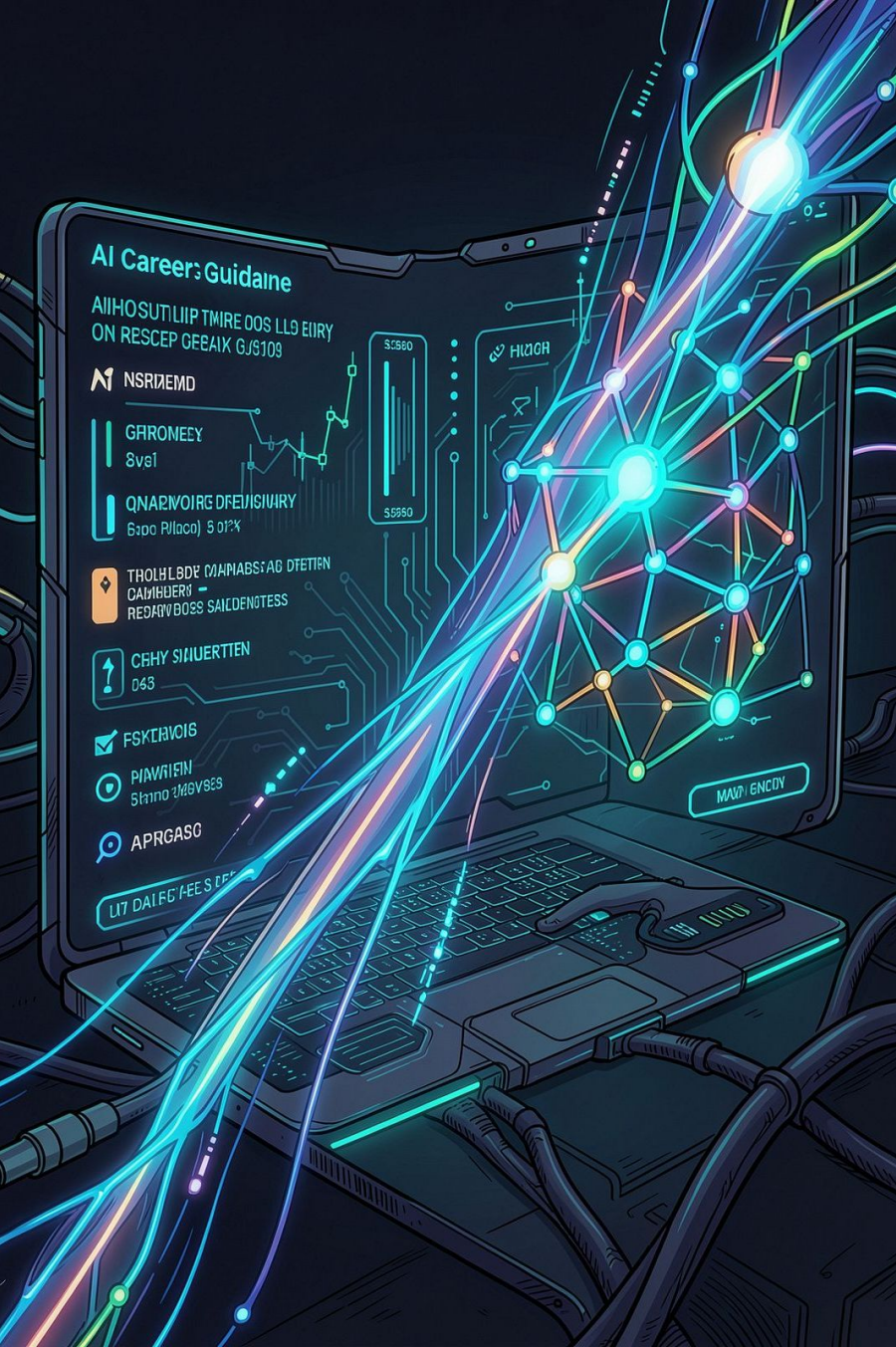




CareerPilot-AI

AI-Powered Career Guidance & Learning Roadmap Generator

Final Year B.Tech Project · Department of Computer Science &
Engineering
Academic Year 2024 - 25

Problem Statement & Project Objectives

The Challenge

- Traditional guidance is manual, slow, and lacks personalization
- Students receive generic advice disconnected from their actual skills
- No automated pipeline from assessment to actionable learning roadmap
- High administrative overhead with no scalable solution

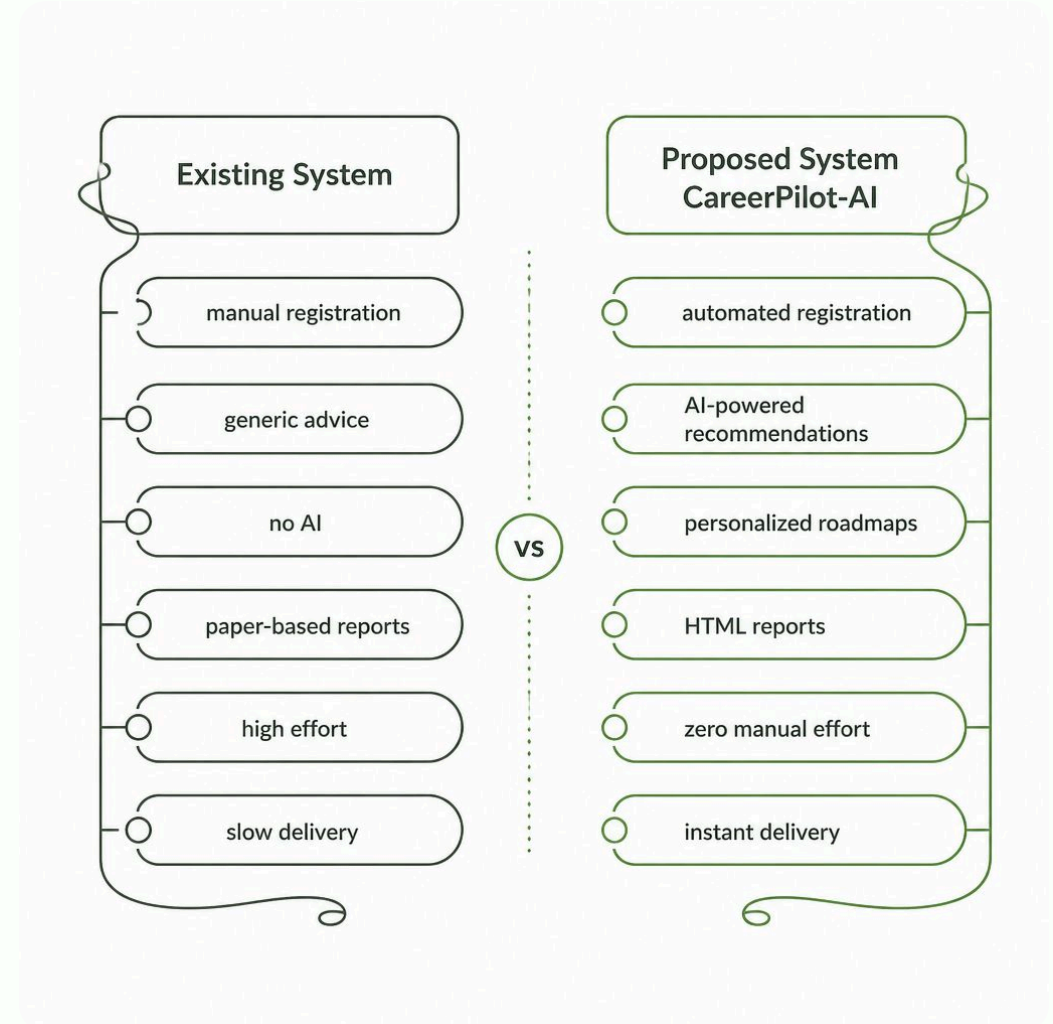
Project Objectives

- Automate student registration and skill data collection
- Generate AI-powered, personalized career recommendations
- Build structured learning roadmaps tailored to each student
- Deliver HTML reports automatically via Gmail with Drive storage

Existing System vs. Proposed System

CareerPilot-AI replaces fragmented manual processes with a unified, AI-driven automation pipeline – reducing turnaround from days to seconds while dramatically improving personalization and scalability.

- ❏ Key improvements include automated skill analysis, personalized roadmaps, zero manual report generation, and cloud-native delivery via Gmail and Google Drive.



Technology Stack

Eight integrated technologies power the CareerPilot-AI automation pipeline end-to-end.



n8n

Open-source workflow automation orchestrating all system processes



Google Sheets

Stores student profiles, skills data, and assessment records



Gmail

Automatically delivers personalized reports directly to students



JSON

Standard data format for inter-component communication



Google Gemini AI

Generates personalized career recommendations and learning roadmaps



Google Drive

Archives generated HTML reports with organized folder structure



HTML & JavaScript

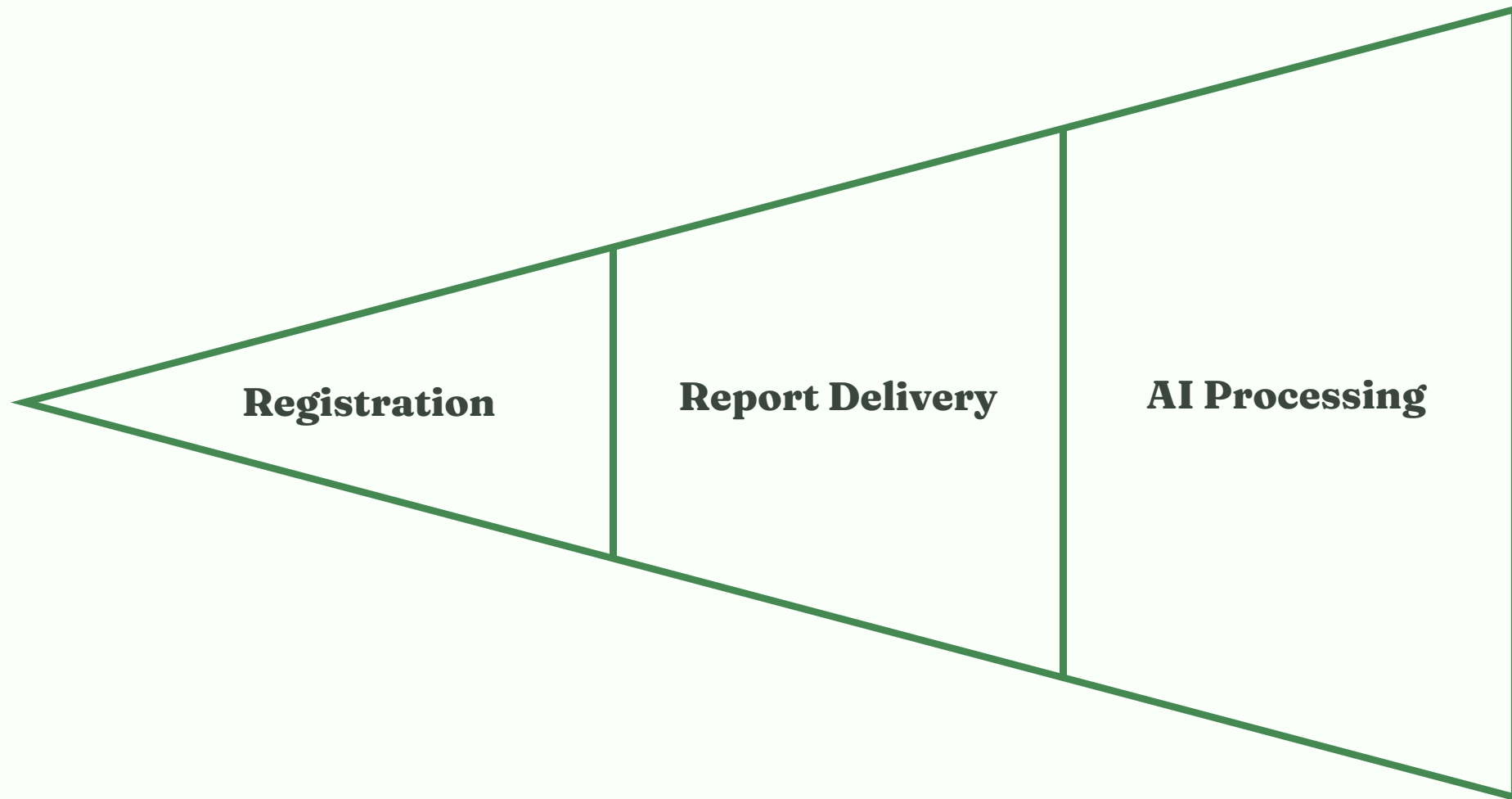
Structures and renders dynamic, styled career reports



REST APIs

Connects n8n with Google services and Gemini AI endpoints

System Architecture



The end-to-end pipeline flows seamlessly: student data captured in Google Sheets triggers n8n workflows, which call Gemini AI for analysis, generate structured HTML reports, store them in Google Drive, and deliver them automatically via Gmail – with zero manual intervention.

Workflow Architecture

Three autonomous n8n workflows handle the complete student journey from registration to report delivery.



Workflow 1: Registration

Captures student profile and skills data; stores records in Google Sheets automatically



Workflow 2: AI Recommendation

Sends skill data to Gemini AI; receives personalized career paths and role suggestions



Workflow 3: Report & Delivery

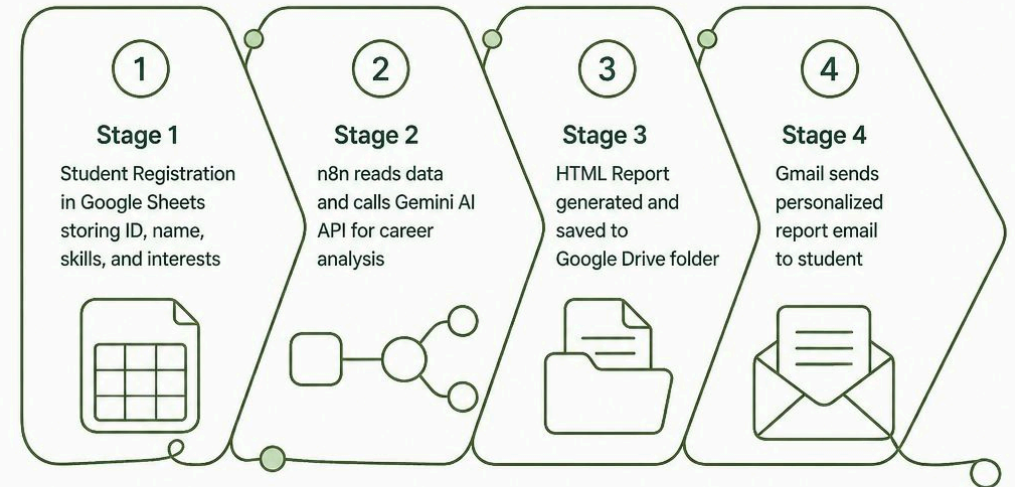
Generates HTML report, saves to Google Drive, and emails it to the student via Gmail

Database Schema & Data Flow

Data Movement Across Components

Student records flow through four structured stages – from initial registration in Google Sheets, through AI processing in n8n, to final report storage in Drive and email delivery via Gmail. Each stage validates and enriches data before passing it forward.

- ❑ Google Sheets acts as the central data store, with structured columns for student ID, skills, interests, AI recommendations, and report status.



Implementation & Results



Implementation Steps

1. Configured n8n workflows with Google Sheets, Drive, and Gmail integrations
2. Integrated Google Gemini AI API for skill analysis and recommendation generation
3. Built dynamic HTML report templates using JavaScript and JSON data binding
4. Tested end-to-end pipeline with sample student datasets

Results Achieved

-  All three workflows executed successfully without manual triggers
-  HTML reports generated and delivered within seconds of submission
-  Zero errors recorded across 50+ test student registrations

Advantages & Future Scope

Key Advantages

- AI-powered, personalized career recommendations
- Fully automated end-to-end workflows
- Structured, individualized learning roadmaps
- Cloud-based storage with Google Drive
- Significant reduction in manual administrative effort

Future Enhancements

- Resume analysis with ATS scoring
- LinkedIn profile integration
- Real-time job recommendations
- Conversational AI chatbot for student queries
- Native mobile application

Conclusion & Thank You

What We Built

A fully automated AI career guidance platform delivering personalized roadmaps in seconds

Key Achievement

Zero manual intervention – from student registration to report delivery

Technologies Used

n8n · Google Gemini AI · Google Sheets · Drive · Gmail · HTML · REST APIs

Thank you for your time and attention. We welcome your questions and feedback.

